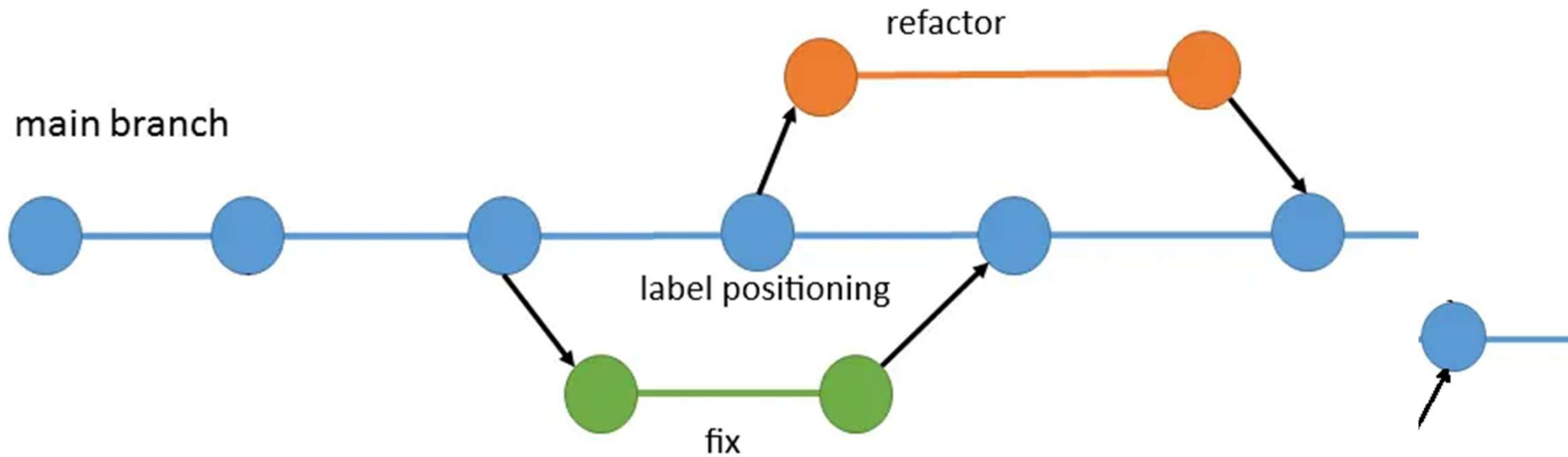




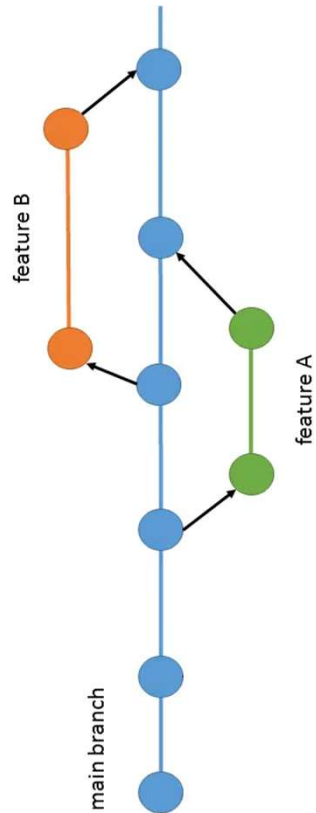
Version Control Basics



Version Control Software

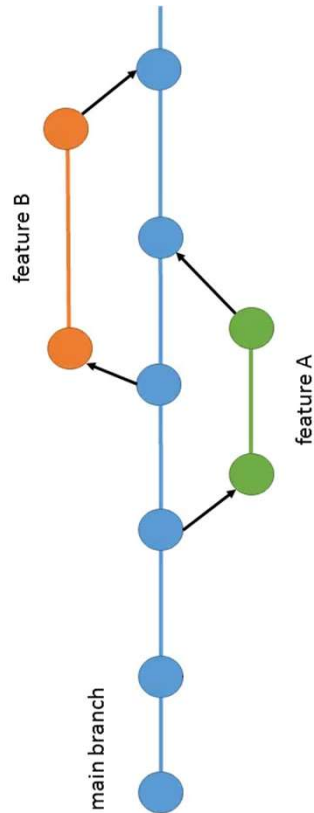


Basic Concepts



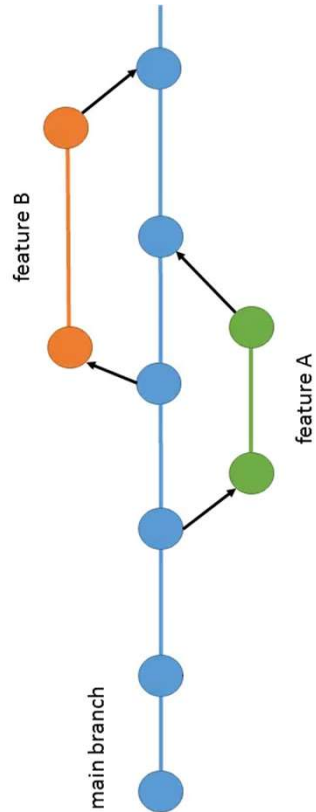
- repository
- branches:
 - default (usually master or main)
 - feature
- fork
- commit, review
- merge, conflict
- revert, rollback

Types of Version Control Software



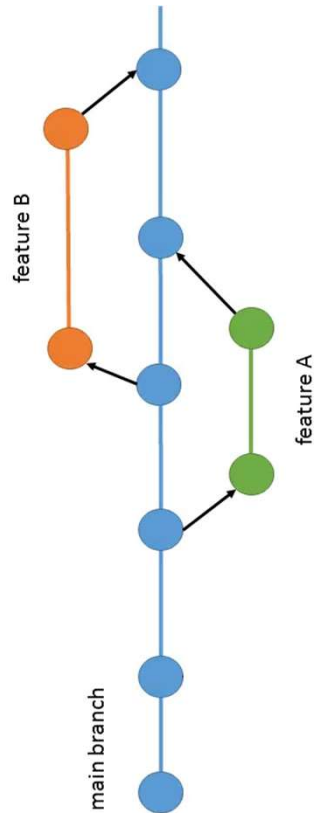
- Based on source code management:
 - Shared
 - All developers work on the same repository
 - Examples: The Librarian (IBM 1969), Panvalet (IBM 1970s)
 - Client-server:
 - Developers work on copies of the project;
 - Changes become available after commit/push
 - History and branches are stored on the server
 - Examples: CVS, SVN
 - Distributed:
 - Developers share projects through a server
 - After download, the full history is available locally, extended with the local changes
 - Version control is possible without network connection
 - Examples: BitKeeper, Git

Git Repository Hosting Services



- GitHub
- SourceForge, GitLab, BitBucket
- AWS Codecommit, Azure ADO

The Most Important Git Commands



- Start
 - clone, init
- Managing changes
 - add, mvn
- Version history
 - log, diff, status
- Branch management
 - branch (create, delete, list, show-current), commit
 - switch, checkout
 - merge, rebase
 - revert, reset
 - tag (create, delete, list)
- Collaboration
 - push, fetch, pull

THANK YOU!